

ZANNATUL FERDOUSEE

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SUMMARY

M.S. student in Computer Science at the University of Minnesota Duluth, conducting research in Human-Robot Interaction, multimodal sensing, and machine learning in the DECRS Lab. Experienced in working with **vulnerable human-subjects** in real-world care settings using physiological and behavioral approach.

RESEARCH INTERESTS

Human-Robot Interaction • Social & Assistive Robotics • Human-Centered Robotics • Multimodal Machine Learning • Physiological & Behavioral Sensing • AI for Healthcare

EDUCATION

MS in Computer Science

University of Minnesota Duluth

GPA: 4.00/4.00

Aug'2025 - present

Minnesota, USA

B.Sc in Computer Science and Engineering

Rajshahi University of Engineering and Technology (RUET)

Jan'2017–Oct'2022

Rajshahi, Bangladesh

RELEVANT GRADUATE COURSEWORK

- Humanoid Robotics: HRI, Kinematics & Autonomy
- Machine Learning & Data Mining
- Sensors & Internet of Things
- Natural Language Processing
- Regression Analysis

SELECTED PUBLICATIONS

- **Zannatul Ferdousee***, Arshia Khan "[Haptics in Robotics: A Systematic Literature Review](#)". In *Studies in Health Technology and Informatics*, 2026, DOI:10.3233/SHTI260905
- **Zannatul Ferdousee***, Varshini Bhavanam, Maggie Beach, Arshia Khan "[Digital Interventions for Depression and Anxiety: A Systematic Synthesis of Chatbots, App-Based CBT, and Immersive VR/XR](#)". In Proceedings of the 19th International Joint Conference on Biomedical Engineering Systems and Technologies (BIOSTEC 2026) – HEALTHINF, pp. 756–761, DOI:10.5220/0014484200004070
- **Zannatul Ferdousee***, Md Rakibul Haque "[Integrating Feature Selection and Extraction with Tuned Multilayer Perceptrons for Predicting Student Dropout Risk in Higher Education](#)". In 2024 2nd International Conference on Information and Communication Technology (ICICT 2024), DOI:10.1109/ICICT64387.2024.10839727
- **Zannatul Ferdousee***, S M Shovan, MD Sharifujjaman "[Propionylation characterization ensembling classifiers on varied features](#)". In 2023 26th International Conference on Computer and Information Technology (ICCIT 2023), DOI:10.1109/ICCIT60459.2023.10441189

THESIS

Master's Thesis (In Progress): *The Role of Touch in Human-Robot Interaction: Affective and Physiological Responses of Older Adults to Social Robot Touch*

- Investigating how robot-initiated touch influences affect, engagement, and physiological responses during human-robot interaction using behavioral measures and EEG/EDA sensing.

WORK EXPERIENCE

- Graduate Teaching Assistant

University of Minnesota Duluth

Aug'2025–Present

 - **Courses:** Finite Mathematics; Introduction to Calculus; Introduction to Computer Science
 - Led discussion and lab sessions, graded coursework, and supported students with course concepts.
- Graduate Research Assistant

University of Minnesota Duluth

Jun'2026 - Aug'2026
Supervisor: Dr. Arshia Khan

 - Conducted a Human-Robot Interaction (HRI) study with older adults at **Duluth Heights Lodge Senior Living**, evaluating engagement, acceptance, and perceived companionship with the Pepper robot through structured interactions and multimodal data collection, including behavioral, EDA, EEG, and survey/interview data.
- Research Assistant

Leaf Technologies

Feb'2023 – Apr'2024
Dhaka, Bangladesh

 - Worked on computer vision and machine learning models for digital signal processing, performing Python-based audio analysis using Fast Fourier Transform (FFT) to extract and evaluate key signal features.

SELECTED PROJECTS

- **ChatGPT-Enabled Conversational System for Pepper Robot:** Integrated ChatGPT and Whisper-based speech recognition with the Pepper humanoid robot to enable natural language interaction through a client-server architecture.
- **Plant Care Robot:** Developed a Raspberry Pi-based robotic system integrating soil-moisture sensing, motor control, servo actuation, and LCD feedback for automated plant monitoring and responsive care.
- **Emotion Recognition in Dialogue:** Developed and evaluated NLP models on the DailyDialog dataset, comparing TF-IDF, DistilBERT, BERT, RoBERTa, and a custom Transformer for multi-class emotion recognition.
- **House Sales Regression Analysis:** Developed & evaluated regression models on the King County housing dataset (21,613 observations), performing model selection, statistical diagnostics, ANOVA, and prediction using R

ACADEMIC ACHIEVEMENTS

- **Technical Scholarship** (Jan'2017–Dec'2022)
Awarded for academic excellence in the Entrance Exam from the Faculty of CSE at RUET RUET

TECHNICAL SKILLS

Languages	C, C++, Python,
ML Frameworks	PyTorch, TensorFlow, Keras, Scikit-learn
ML Tools	OpenCV, NumPy, Pandas, Matplotlib
Databases	MySQL
Software & Tools	LaTeX, MS Office

EXTRA-CURRICULAR ACTIVITIES

- **SSF Committee Member:** UMD Jan'2026- Present
Reviewed proposals and helped allocate approximately **\$53M** in student service funding annually.
- **Student Mentor:** Dept of CS,UMD Aug'2026- Present
Mentor first-year M.S. students in academics, research, technical writing, and presentation preparation.
- **Leadership:** Class Representative, Dept of CSE, RUET 2017-2020
Acted as a liaison between students and faculty, facilitating communication, coordinating exam schedules, and representing the CSE department at official events and meetings.